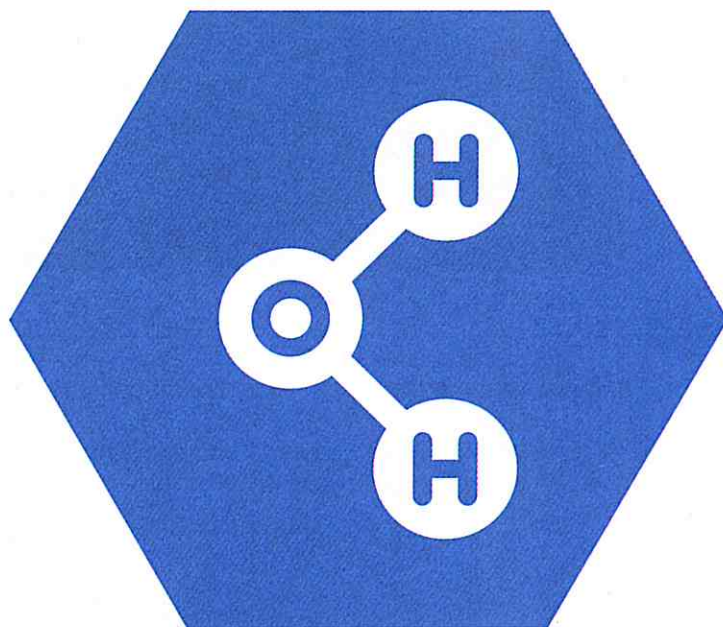




**UNSW Global**  
THE UNIVERSITY OF NEW SOUTH WALES  
SYDNEY • AUSTRALIA

# PAPER C



# 2013 ICAS

International Competitions  
and Assessments for Schools

## SCIENCE

**DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.**

**40 QUESTIONS**

**TIME ALLOWED: 55 MINUTES**

**STUDENT'S NAME:**

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.  
Use a 2B or B pencil.  
Do **NOT** use a pen.  
Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

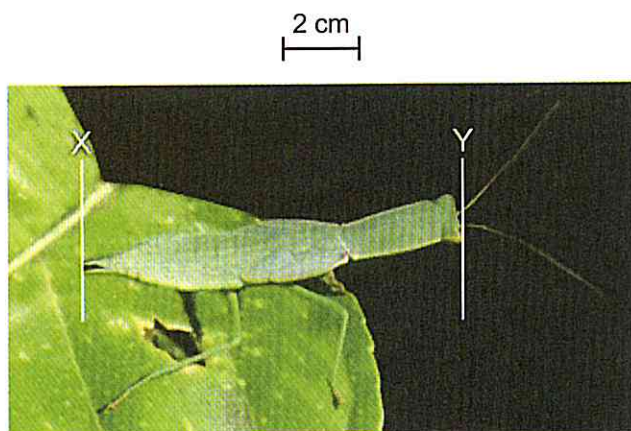
Mark only **ONE** answer for each question.  
Your score will be the number of correct answers.  
Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).  
Use the information provided to choose the **BEST** answer from the four possible options.  
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. The praying mantis is an insect.

The photograph shows a praying mantis on a leaf.

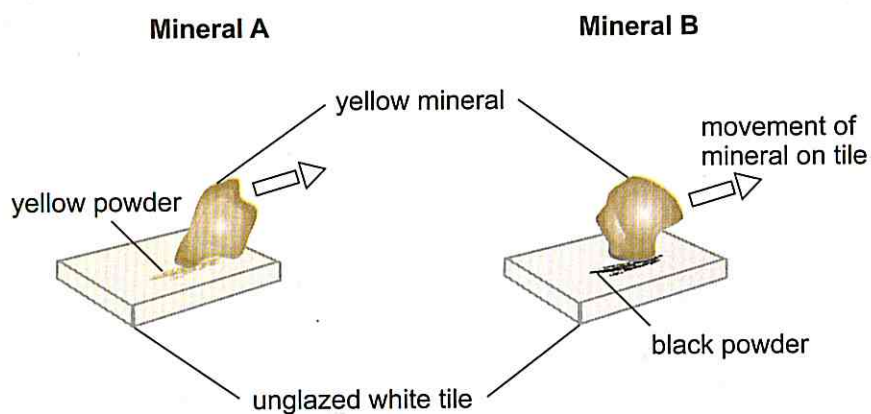


What is the length (in cm) of the praying mantis from X to Y?

- (A) 2
- (B) 4
- (C) 5
- (D) 10

2. Minerals can be distinguished by their properties such as lustre (shine) and streak.

When a mineral is scratched on a rough surface that is harder than itself, a powder is left behind. The colour of the powder is called the streak of the mineral.



Using only the diagram, which property best distinguishes Mineral A from Mineral B?

- (A) lustre
- (B) colour
- (C) streak
- (D) hardness

3. For a rocket to be launched and leave a planet, it must reach a certain speed. This speed is called the escape velocity.

The table shows the escape velocity of each planet in the solar system.

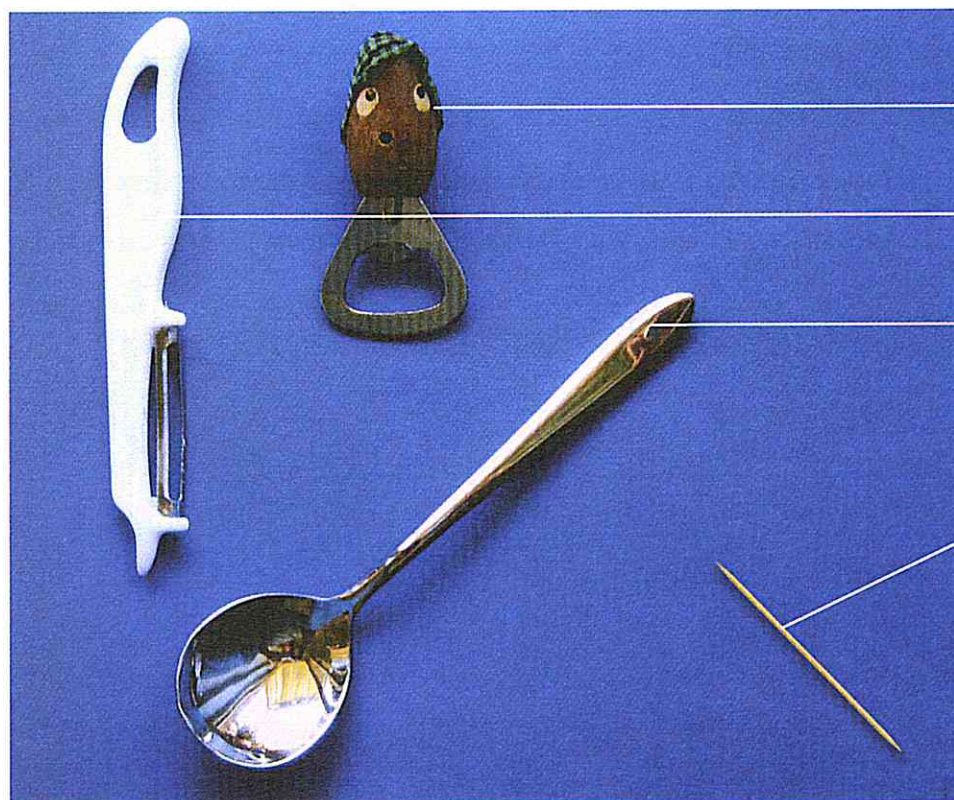
| Planet  | Escape velocity (km/s) |
|---------|------------------------|
| Mercury | $4 \frac{1}{2}$        |
| Venus   | $10 \frac{1}{2}$       |
| Earth   | 11                     |
| Mars    | 5                      |
| Jupiter | $59 \frac{1}{2}$       |
| Saturn  | $35 \frac{1}{2}$       |
| Uranus  | $21 \frac{1}{2}$       |
| Neptune | $23 \frac{1}{2}$       |

How many planets have an escape velocity greater than that of Earth?

- (A) 3  
(B) 4  
(C) 5  
(D) 6

For questions 4 and 5 use the information below.

Jack took a photograph of some objects he found in the kitchen.



bottle opener

vegetable peeler

spoon

toothpick

4. Which object is the longest?
- (A) bottle opener      (B) spoon      (C) vegetable peeler      (D) toothpick
5. Which of the following would **NOT** be helpful when ordering the objects by length?
- (A) placing them close together      (B) placing them next to a ruler  
(C) lining them up the same way      (D) weighing each object

For questions 6 and 7 use the information below.

The diagram represents some craters on the surface of the Moon.

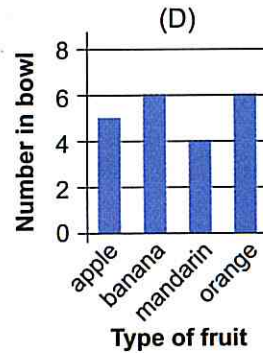
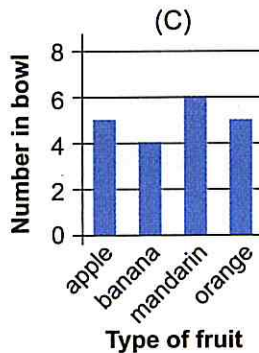
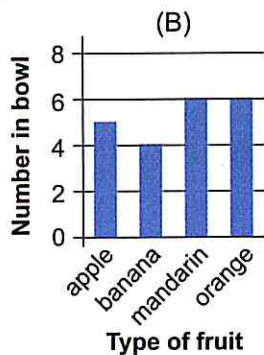
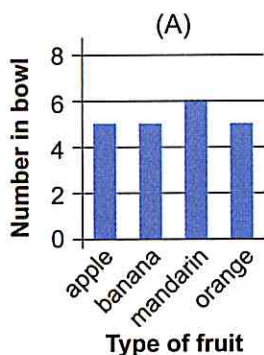


6. Which statement is correct?
- (A) Crater R is wider than crater T.  
 (B) Crater U is wider than crater S.  
 (C) Crater T is wider than crater Q.  
 (D) Crater Q is wider than crater R.
7. Which statement is supported by the information in the diagram?
- (A) Crater R was formed before crater S.  
 (B) Crater Q was formed before crater R.  
 (C) Crater V was formed before crater T.  
 (D) Crater U was formed before crater V.

8. Jason counted how many pieces of fruit there were in the fruit bowl in the dining room.  
 The table shows his results.

| Type of fruit | Number in bowl |
|---------------|----------------|
| apple         | 5              |
| banana        | 4              |
| mandarin      | 6              |
| orange        | 5              |

Which graph shows the information in Jason's table?

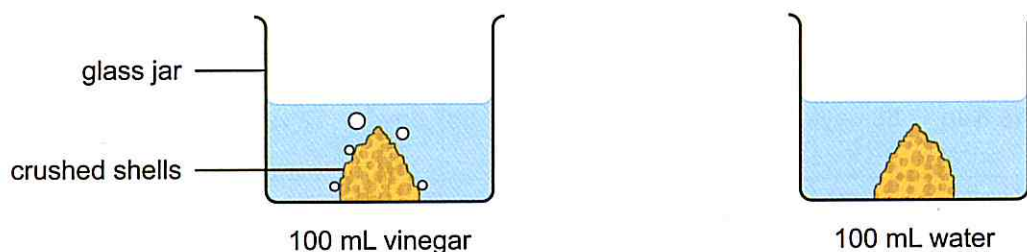


For questions 9 and 10 use the information below.

Simon predicted that seashells would dissolve in vinegar but not in water.

He set up an investigation using crushed shells.

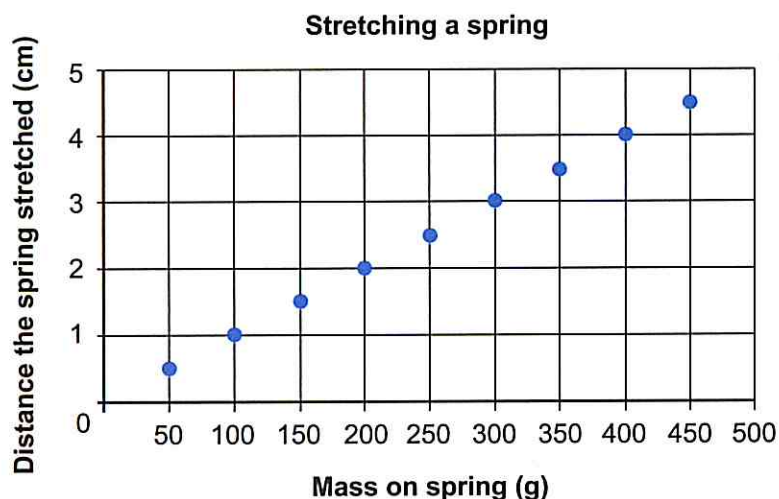
Simon drew these two diagrams to show his teacher what happened in his investigation.



9. What did Simon have to do to make this a fair test?
- (A) Fill both jars to the same level. (B) Use the same liquid in both jars.  
(C) Use jars that have the same shape. (D) Keep the two jars at the same temperature.
10. Which of the following is an observation that Simon recorded in his diagrams?
- (A) The vinegar became warm.  
(B) The crushed shells dissolved in the vinegar.  
(C) The crushed shells in both jars made bubbles.  
(D) The crushed shells in the vinegar made bubbles.

11. A spring stretches when you hang a mass from it.

James attached different masses to the bottom of a spring and measured how far the spring stretched in each case. The graph below summarises his results.



What mass is needed to make the spring stretch exactly 2 cm?

- (A) 200 g  
(B) 300 g  
(C) 400 g  
(D) 500 g

12. The table provides information about some of the planets in our solar system.

|                                 | Mercury | Venus    | Earth    | Mars     | Jupiter  | Saturn   | Uranus   |
|---------------------------------|---------|----------|----------|----------|----------|----------|----------|
| Diameter (km)                   | 4 879   | 12 104   | 12 756   | 6 792    | 142 984  | 120 536  | 51 118   |
| Distance from Sun (millions km) | 58      | 108      | 150      | 228      | 779      | 1433     | 2872     |
| Time to go around the Sun once  | 88 days | 225 days | 365 days | 687 days | 12 years | 29 years | 84 years |
| Time to spin on axis once       | 59 days | 243 days | 24 hours | 25 hours | 10 hours | 11 hours | 17 hours |

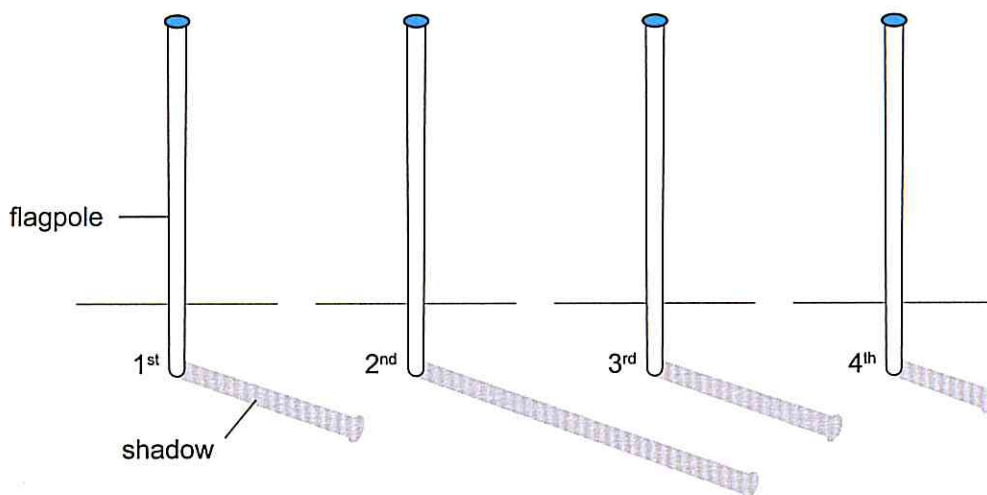
Which two planets are most similar in size?

- (A) Mercury and Mars  
(B) Venus and Earth  
(C) Earth and Mars  
(D) Jupiter and Saturn

13. The apparent path of the Sun across the sky changes with the seasons. The Sun stays closer to the horizon during winter and rises highest in the sky during summer.

Joshua drew the shadow made by a flagpole on four days, three months apart, and always at the same time of day.

The diagram shows his drawings in the order in which he drew them.



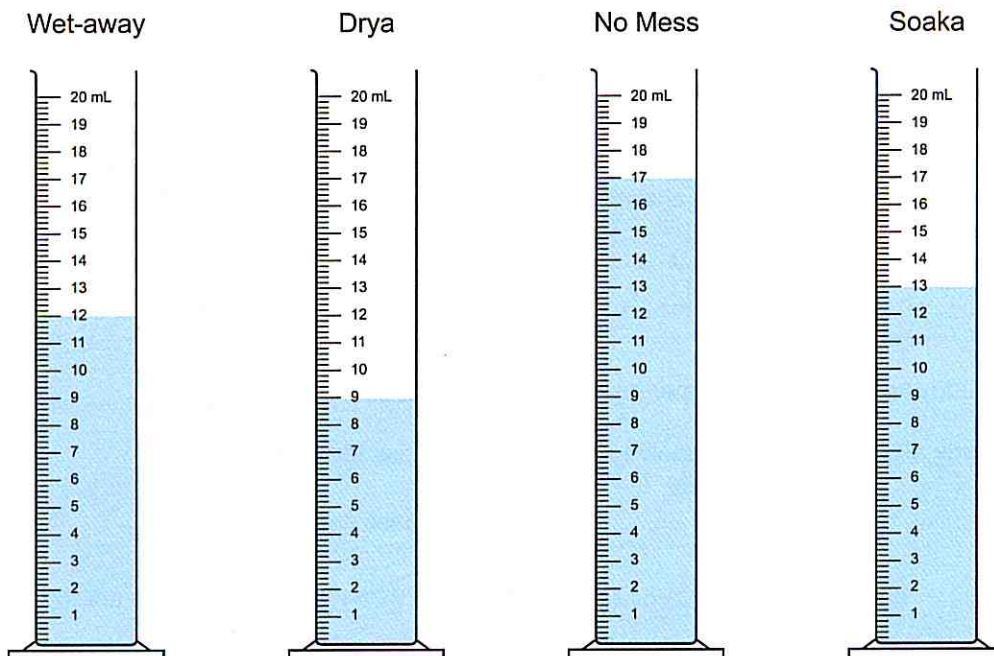
In which season did he make the first drawing?

- (A) autumn (B) summer (C) winter (D) spring

For questions 14 and 15 use the information below.

Sharon and Andrew wanted to compare four different brands of paper towel. They had equal-sized paper towels of four different brands: Wet-away, Drya, No Mess, Soaka. They had four measuring cylinders, one for each paper towel. Each measuring cylinder contained 20 mL of water.

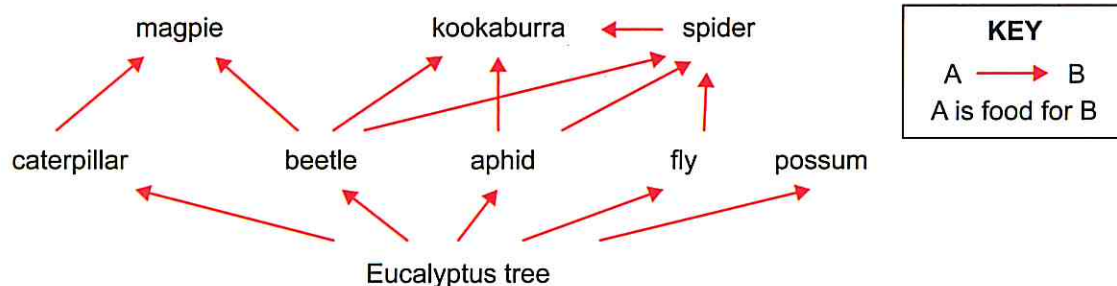
Sharon and Andrew slowly poured water onto each paper towel until the paper towel could not soak up any more. The diagrams show the water left in the measuring cylinders after they had stopped pouring water onto the paper towels.



14. Was this a fair test?
- (A) No, because each paper towel soaked up different amounts of water.
  - (B) Yes, because they used the same amount of water each time.
  - (C) No, because the paper towels were all different brands.
  - (D) Yes, because each paper towel was the same size.
15. What were Sharon and Andrew trying to find out?
- (A) Are all paper towels the same size?
  - (B) Will paper towels soak up different liquids?
  - (C) Which liquid do the paper towels soak up best?
  - (D) Do all paper towels soak up the same amount of water?

16. Plants are called producers because they make food using energy from the Sun. An organism that eats another organism is called a consumer. A consumer that eats a producer is called a first-order consumer.

This diagram shows a food web.

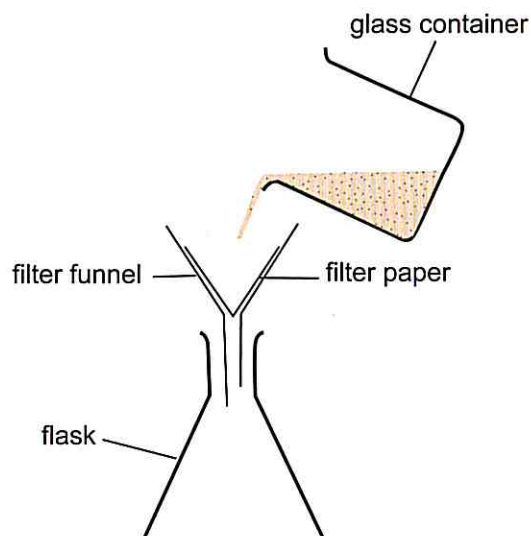


How many first-order consumers are shown in this food web?

- (A) 1                      (B) 3                      (C) 4                      (D) 5

17. Ali poured muddy water onto filter paper inside a filter funnel.

Filter paper only allows water and dissolved substances to pass through it.



Which option correctly describes the contents of the filter paper and the contents of the flask after Ali finished filtering?

|     | Contents of filter paper | Contents of flask |
|-----|--------------------------|-------------------|
| (A) | mud                      | clear water       |
| (B) | muddy water              | muddy water       |
| (C) | empty                    | clear water       |
| (D) | clear water              | mud               |

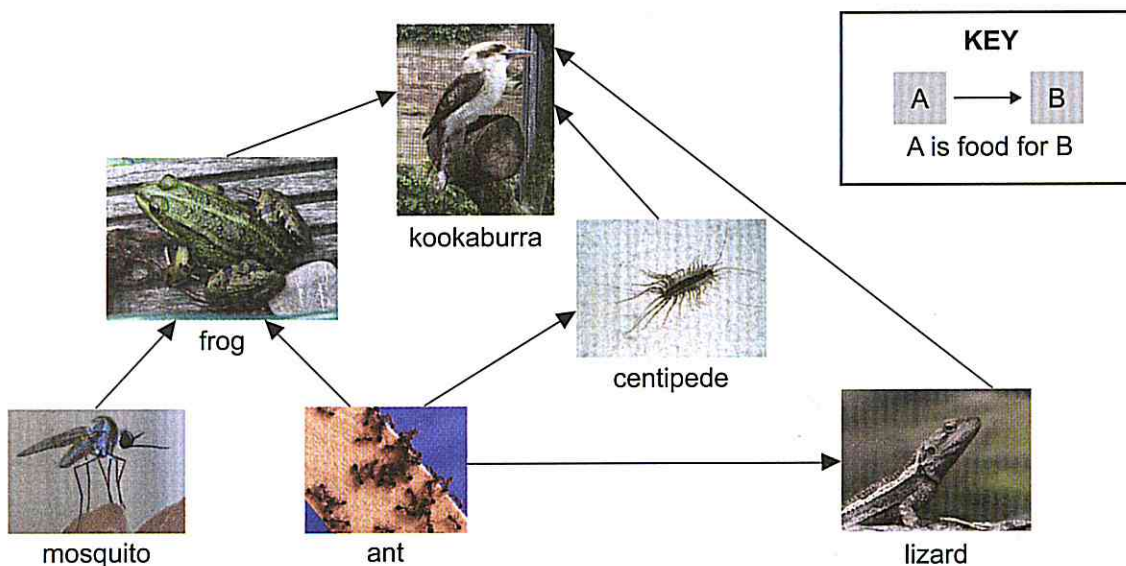
18. Liam has a rock collection. He uses this key to identify his rocks.

1. Fizzes when acid is poured on it .....limestone  
Does not fizz when acid is poured on it .....Go to 2
2. Floats on water .....pumice  
Does not float on water.....Go to 3
3. Black, glassy appearance .....obsidian  
Particles (grains or crystals) are visible.....Go to 4
4. Made of grains that are easy to rub off .....sandstone  
Made of crystals that are tightly bound together.....Go to 5
5. Crystals are in layers .....gneiss  
Crystals are not in layers .....Go to 6
6. Small dark coloured crystals .....basalt  
Larger light coloured crystals .....granite

Which is the best description of gneiss?

- (A) fizzes when acid is added, composed of visible particles, floats on water
- (B) not composed of visible particles, does not float on water, small dark crystals
- (C) crystals arranged in layers, does not float on water, does not fizz when acid is added
- (D) not composed of visible particles, has a black, glassy appearance, crystals arranged in layers

19. This food web shows the feeding relationships between some animals living in a given area.



What is likely to happen if a disease kills most of the ants living in this area?

- (A) Fewer lizards will survive.
- (B) The frogs will have no food.
- (C) There will be more mosquitoes.
- (D) The kookaburras will have more food to eat.

For questions 20 and 21 use the information below.

Different animals have different hearing ranges. An animal's lower limit of hearing is the lowest frequency sound it can hear. Its upper limit of hearing is the highest frequency sound it can hear. The hearing range of an animal is made up of all the sound frequencies in between.

The table shows information about some animals.

| Animal       | Lower limit of hearing (Hz) | Upper limit of hearing (Hz) |
|--------------|-----------------------------|-----------------------------|
| bird         | 250                         | 21 000                      |
| crocodile    | 20                          | 6 000                       |
| dog          | 15                          | 50 000                      |
| bat          | 1 000                       | 150 000                     |
| African moth | 5 000                       | 110 000                     |
| dolphin      | 150                         | 150 000                     |

20. How many of these animals can hear a sound with a frequency of 125 Hz?

- (A) 0                      (B) 1                      (C) 2                      (D) 3

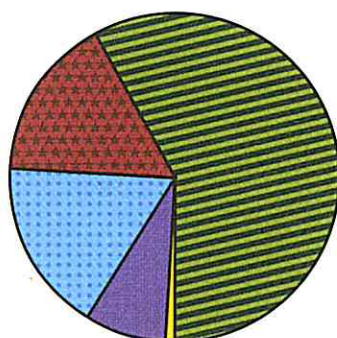
21. A siren makes a noise that both the bat and the crocodile are able to hear.

In what range must the frequency of the siren noise lie?

|     | Lowest frequency (Hz) | Highest frequency (Hz) |
|-----|-----------------------|------------------------|
| (A) | 20                    | 1 000                  |
| (B) | 20                    | 6 000                  |
| (C) | 1 000                 | 6 000                  |
| (D) | 6 000                 | 150 000                |

22. There are several different gases that cause the greenhouse effect. Greenhouse gases trap heat in the atmosphere.

The chart shows the proportions of some types of greenhouse gases in the atmosphere.



**KEY**

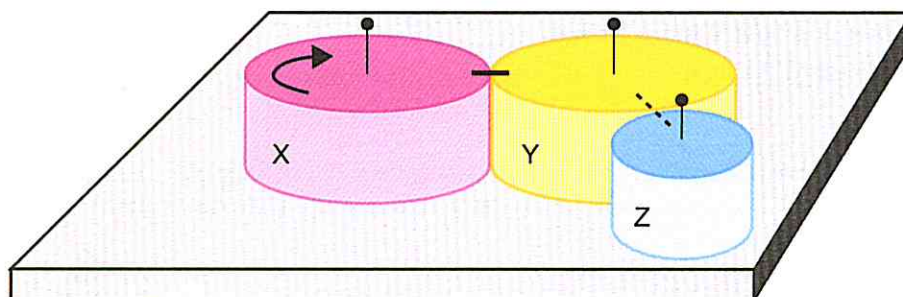
-  carbon dioxide from fossil fuels
-  halocarbons
-  nitrous oxide
-  methane
-  carbon dioxide from deforestation

Approximately what percentage of these greenhouse gases in the atmosphere consists of carbon dioxide?

- (A) 75%                      (B) 60%                      (C) 40%                      (D) 15%

23. Fiona built a three-wheeled gear chain by connecting the gears by their outer edges (rims). She used two identically sized wheels and one wheel with half the diameter of the other two.

Fiona made a solid mark where wheel X touched wheel Y and a dotted mark where wheel Y touched wheel Z. She made sure that when she turned one wheel the other two wheels also turned without slipping.



Fiona turned wheel X **two** times clockwise in the direction shown on the diagram.

How many times will wheel Z turn, and in what direction?

| Wheel Z |                 |                     |
|---------|-----------------|---------------------|
|         | Number of turns | Direction of motion |
| (A)     | 4               | clockwise           |
| (B)     | 1               | clockwise           |
| (C)     | 4               | anticlockwise       |
| (D)     | 1               | anticlockwise       |

24. The table shows the approximate speed at which light and sound travel in different substances.

| Substance | Speed of light (m/s) | Speed of sound (m/s) |
|-----------|----------------------|----------------------|
| air       | 300 000 000          | 340                  |
| water     | 225 000 000          | 1 484                |
| glass     | 197 000 000          | 5 960                |

Based on the information in the table, which statement is correct?

- (A) Light slows down as it passes from water to glass.
- (B) Light travels at the same speed through all substances.
- (C) Sound speeds up as it passes from glass to water.
- (D) Sound travels faster than light in some substances.

25. Ann carried out an experiment to determine how long it took to boil salt water.

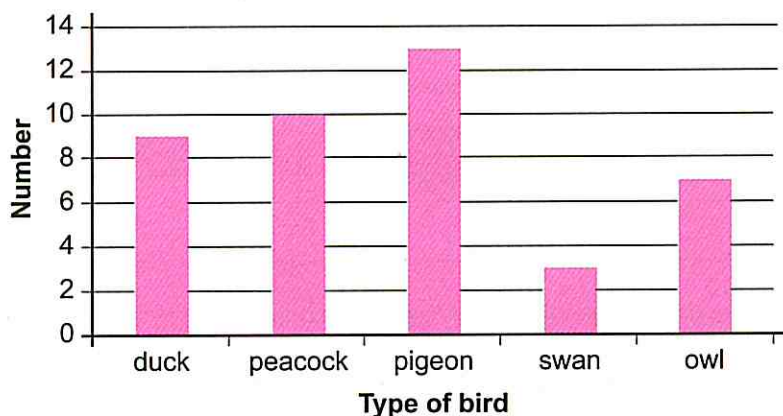
The steps of the procedure are listed but they are not in the correct order:

1. Heat the beaker using a Bunsen burner until bubbles form in the water.
2. Weigh 5 grams of salt and put it in the beaker.
3. Measure 150 mL of water and pour it into a beaker.
4. Stop the stopwatch and record the time.
5. Start the stopwatch.

In which order should Ann carry out these steps?

- (A) 3, 2, 5, 4, 1  
(B) 3, 2, 5, 1, 4  
(C) 5, 3, 1, 2, 4  
(D) 5, 2, 3, 4, 1

26. Isabel went to the zoo with her parents. She recorded the numbers of different types of birds that she saw. She used her results to draw a graph of her observations.

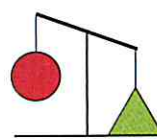
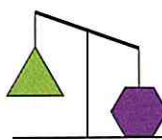
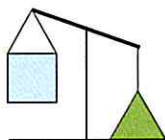
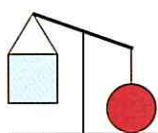


How many more peacocks than owls did she see?

- (A) 3                      (B) 7                      (C) 10                      (D) 13

27. Tina has four different shaped objects. She compared them, two at a time, using a balance to see which object was heavier.

The diagrams show the results of her investigation.

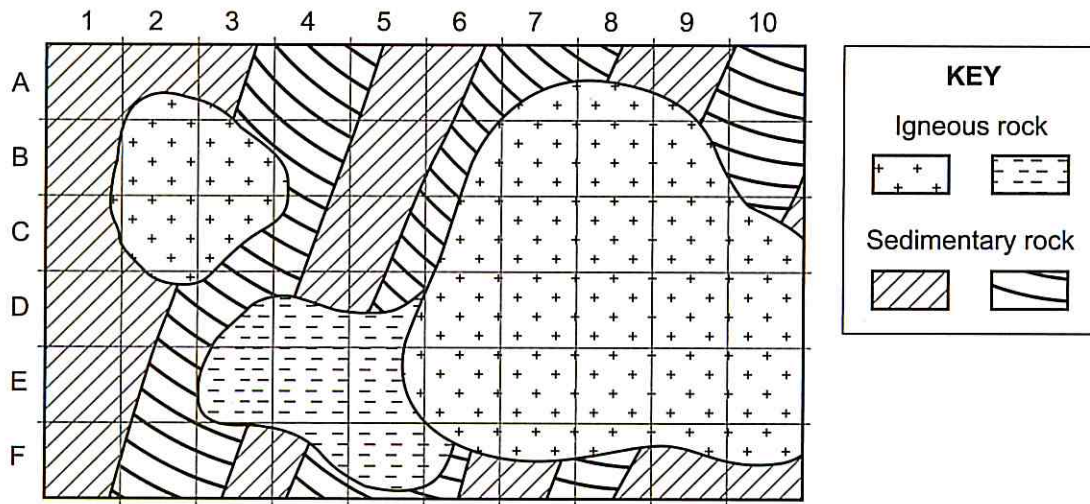


Which object was the heaviest?

- (A)                       (B)                       (C)                       (D) 

28. This map shows the types of rock found in a particular region.

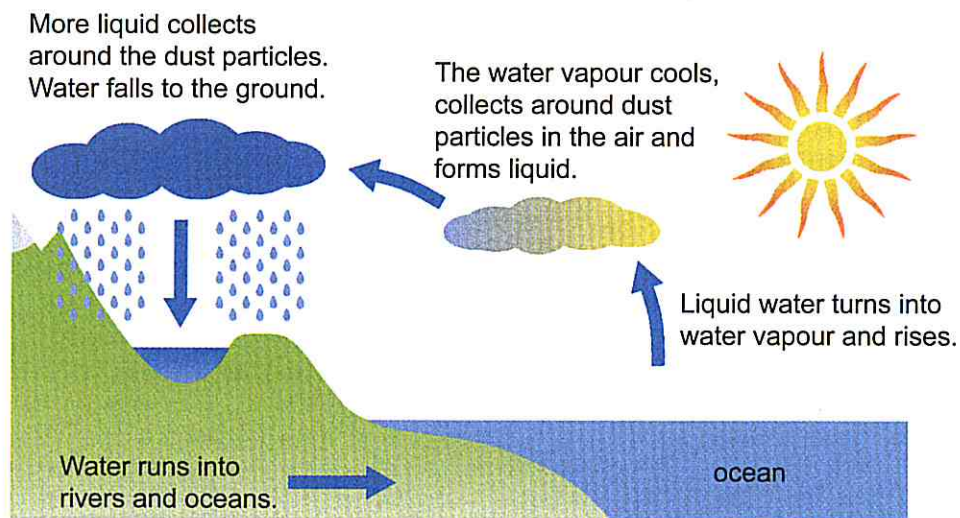
Deposits of quartz are often located at the boundary of igneous and sedimentary rock. Gold is sometimes found in rock containing quartz.



Which of the following locations is most likely to contain gold?

- (A) A6 (B) B2 (C) C3 (D) E5

29. This diagram of the water cycle shows how water changes and moves in, on and around the Earth.



Which of these supplies the energy for the water cycle?

- (A) Sun (B) rain (C) ocean (D) cloud

30. The table shows the average distance from the Sun of four planets, and the approximate time it takes each planet to go around the Sun once.

| Planet  | Distance from the Sun<br>(millions of kilometres) | Time to go around the Sun<br>(Earth days) |
|---------|---------------------------------------------------|-------------------------------------------|
| Mercury | 58                                                | 88                                        |
| Venus   | 108                                               | 225                                       |
| Earth   | 150                                               | 365                                       |
| Mars    | 228                                               | 687                                       |

Which statement is supported by the information in the table?

- (A) Mars travels at a greater speed than Venus.  
(B) Venus travels at a greater speed than Mercury.  
(C) Mars takes more time to go around the Sun than Earth does.  
(D) Venus takes less time to go around the Sun than Mercury does.

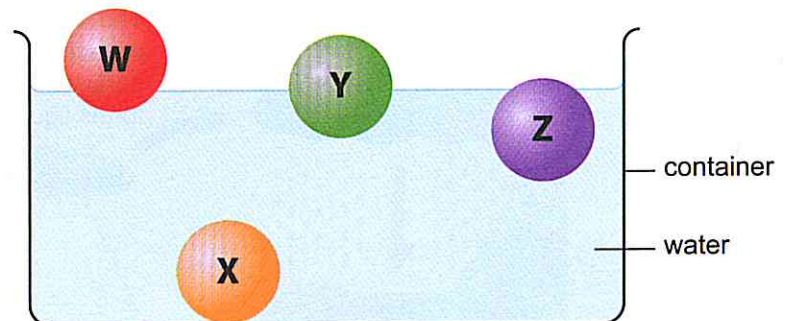
31. Objects that have a density greater than that of water sink in water. Objects that have a density less than that of water float on water. The lower the density of the object, the higher it will float.

Sam had four solid spheres, each one made from a different type of wood. She placed the wooden spheres into a container with water.

The table shows the density of each type of wood. The density of water is  $1.0 \text{ g/cm}^3$ .

The diagram shows what she observed.

| Type of wood | Density ( $\text{g/cm}^3$ ) |
|--------------|-----------------------------|
| balsa        | 0.14                        |
| box          | 1.10                        |
| gum          | 0.95                        |
| maple        | 0.53                        |

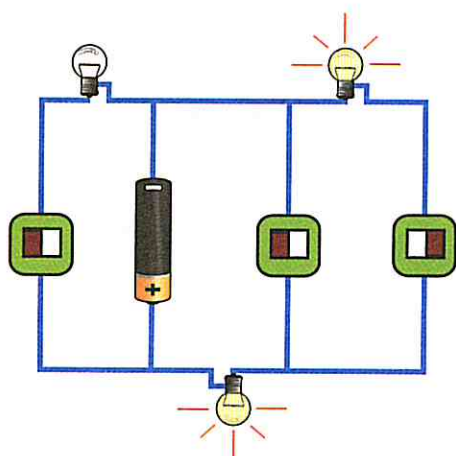


Which sphere was made of maple wood?

- (A) W                      (B) X                      (C) Y                      (D) Z

32. Electricity can flow in a circuit when there is an unbroken path for the electricity to follow. An open switch breaks the path.

The diagram shows an electric circuit. All of the globes have been tested and they work.



#### KEY DIAGRAM

 dry cell

 globe not glowing

 globe glowing

 switch open

 switch closed

#### KEY CIRCUIT

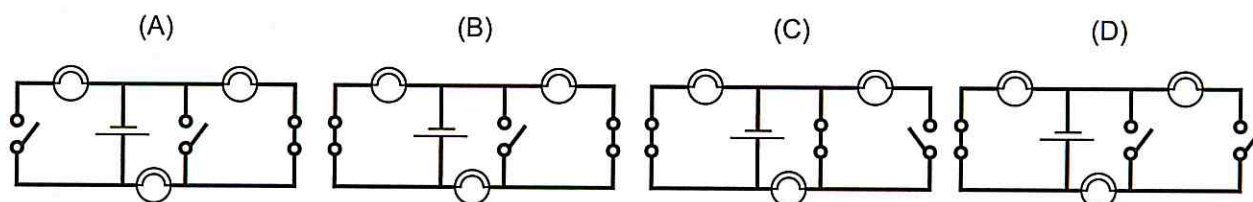
 dry cell

 globe

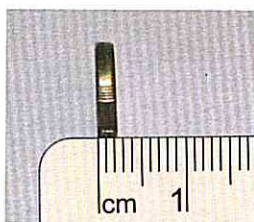
 switch open

 switch closed

Which circuit matches the diagram shown above?



33. Jason used a ruler to measure the thickness of a coin. He recorded his measurement as "about two millimetres". Jason's teacher asked him to make a more accurate estimate.



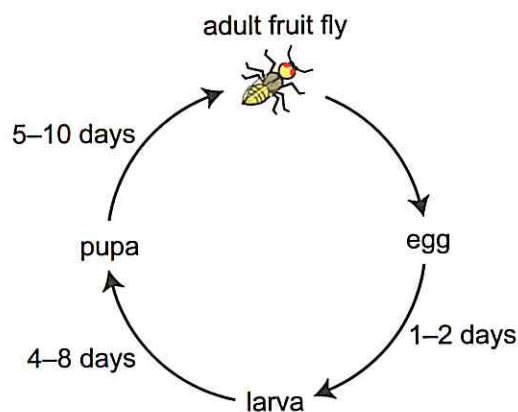
Which of the following is the best way to get a more accurate estimate of the thickness of the coin?

- (A) use a magnifying glass to look more closely at the scale on the ruler
- (B) measure the thickness of the coin with ten different rulers and average the measurements
- (C) measure the total thickness of 10 of the same type of coin placed side by side and divide by 10
- (D) measure the thickness of 10 of the same type of coin separately and average the measurements

For questions 34 and 35 use the information below.

The life cycle of an insect is the process of development from when an egg is laid to the time the insect is able to reproduce.

The diagram shows the life cycle of a fruit fly, and gives some facts about fruit flies.



adult fruit flies live 8–10 days and produce 400–500 eggs

fruit flies lay eggs as soon as they become adults

eggs are inserted into the flesh of ripe fruit by female fruit flies

larvae eat fruit

larvae leave fruit in order to pupate

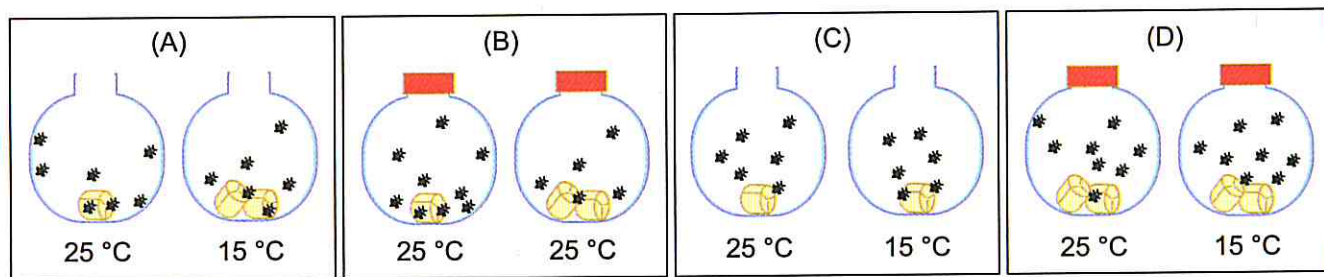
adult fruit flies emerge from the pupae

34. Which is a conclusion that could be drawn from this information?

Fruit flies

- (A) give birth to live young.  
 (B) can reproduce in less than two weeks.  
 (C) spend more than half their life eating fruit.  
 (D) change to adults during the larval stage of their development.
35. Simone read that temperature affected the length of the life cycle of fruit flies. She decided to investigate this idea. She trapped some fruit flies and placed them, together with pieces of banana, in glass jars. She observed the fruit flies for two weeks.

Which setup would give Simone a fair test of the effect of temperature on the life cycle of a fruit fly?

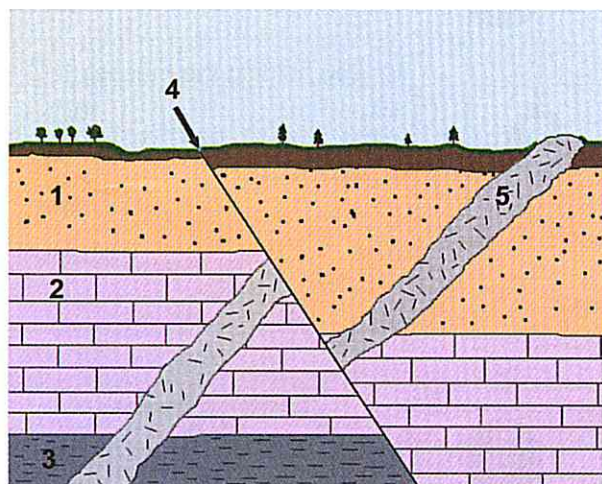


36. The Law of Superposition states that layers of rock in an undisturbed sequence of rocks are arranged in a time sequence, with the oldest layer at the bottom and the newest layer on top.

The diagram shows three rock layers 1, 2 and 3.

The diagram also shows a fault line, 4, where earthquake activity caused the rocks to move past each other.

The diagram shows an intrusion, 5, where molten rock from deep inside Earth pushed its way through the layers towards the surface.



In which order were the different rock layers, the fault and the intrusion formed, from oldest to youngest?

- (A) 1, 3, 2, 4, 5  
 (B) 3, 2, 1, 4, 5  
 (C) 3, 2, 1, 5, 4  
 (D) 5, 4, 1, 2, 3

37. Bananas change colour from green to yellow as they ripen.

All fruit produces ethylene gas when ripening. This gas speeds up the ripening of the fruit.

Tom wanted to see if he could make a banana ripen faster.

He took four plastic bags and placed bananas in them as shown in the diagram.

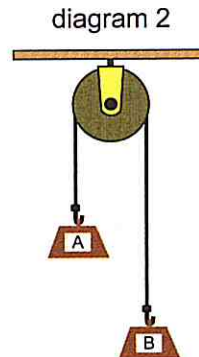
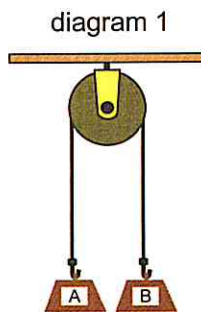


Which bags must Tom choose to make his investigation a fair test and for what reason?

|     | Bags    | Reason                                                                           |
|-----|---------|----------------------------------------------------------------------------------|
| (A) | 1 and 3 | to see which banana ripens first                                                 |
| (B) | 2 and 3 | to see which banana ripens first                                                 |
| (C) | 1 and 4 | to see if a ripening banana makes a difference to the ripening of a green banana |
| (D) | 2 and 4 | to see if a ripening banana makes a difference to the ripening of a green banana |

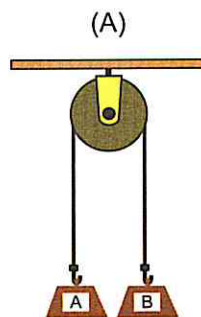
38. Ms Lee had two identical masses (A and B) joined by a light piece of string. She passed the string over a pulley so that the masses could be easily moved up and down. She held the two masses still in the position shown in diagram 1. When she let go of them, they stayed in that position.

Then Ms Lee moved the masses to the position shown in diagram 2 and held them still again.

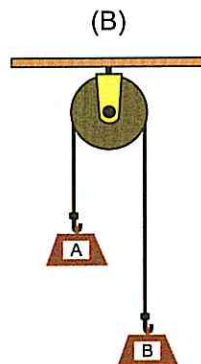


Ms Lee holding the masses.

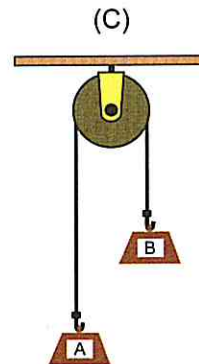
What will happen when Ms Lee lets go of the two masses?



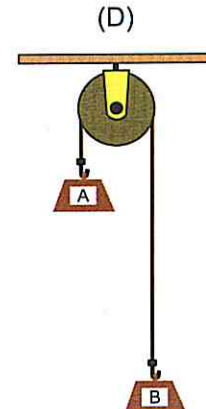
The masses move back to where they were originally and stop.



The masses stay where they are.



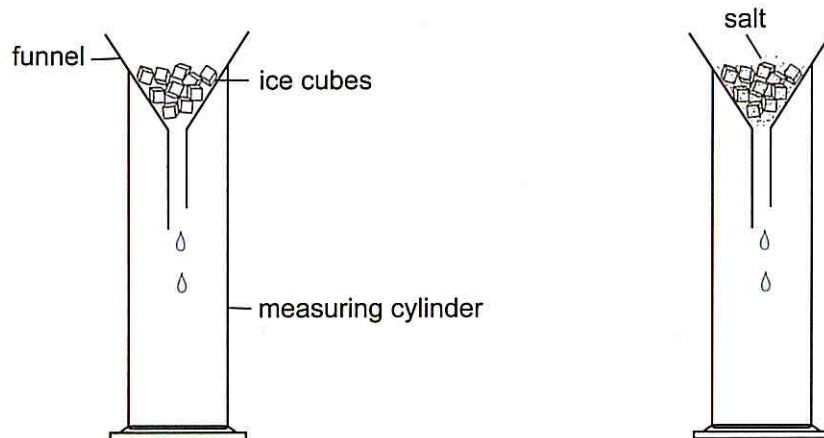
The masses move back and past where they were originally.



The masses move further past where they were moved to.

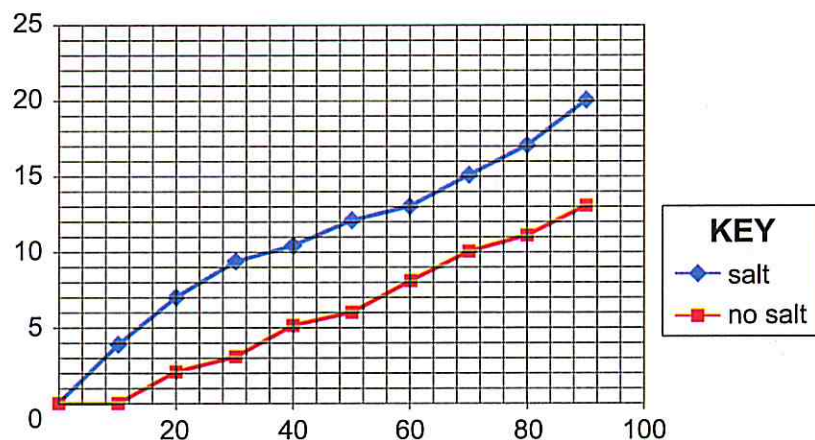
For questions 39 and 40 use the information below.

Lana investigated the effect of salt on melting ice. She placed equal amounts of ice in two funnels. She added salt to the ice in one funnel.



She placed the measuring cylinders close together in the classroom and measured the amount of water in each cylinder at 10-minute intervals.

The graph represents some of her results.



39. Which of the following shows the correct labels for her graph?

|     | Title                         | Horizontal axis  | Vertical axis        |
|-----|-------------------------------|------------------|----------------------|
| (A) | Time for ice to melt          | Time (minutes)   | Amount of salt (g)   |
| (B) | Effect of salt on melting ice | Time (minutes)   | Amount of water (mL) |
| (C) | Salt on ice                   | Ice and salt     | Time (minutes)       |
| (D) | Time for salt to dissolve     | Melting ice (mL) | Time (minutes)       |

40. What can Lana conclude from her results?

- |                                    |                                                 |
|------------------------------------|-------------------------------------------------|
| (A) Ice needs salt to melt.        | (B) Salt makes ice melt faster.                 |
| (C) Ice melts faster without salt. | (D) The more salt the faster the ice will melt. |

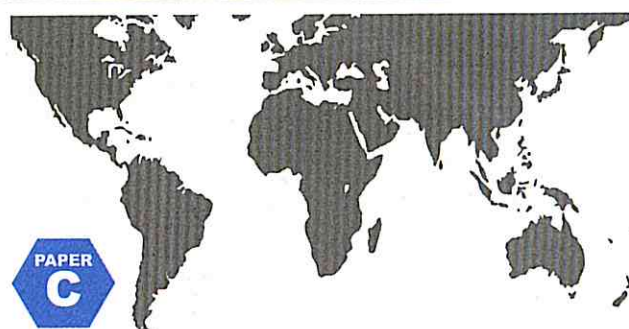
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| Pacific      | Year 6     |
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| South Africa | Grade 5    |



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S

# 2013 Science Answer Keys

ICAS

International Competitions  
and Assessments for Schools

| Question Number | Paper A | Paper B | Paper C | Paper D | Paper E | Paper F | Paper G | Paper H | Papers I & J |
|-----------------|---------|---------|---------|---------|---------|---------|---------|---------|--------------|
| 1               | D       | D       | D       | B       | C       | B       | D       | C       | C            |
| 2               | C       | D       | C       | C       | A       | D       | D       | A       | D            |
| 3               | D       | B       | B       | C       | D       | C       | B       | B       | A            |
| 4               | B       | A       | B       | B       | B       | B       | B       | B       | B            |
| 5               | D       | D       | D       | C       | D       | C       | C       | A       | B            |
| 6               | C       | D       | C       | B       | D       | B       | A       | C       | A            |
| 7               | A       | A       | A       | D       | B       | A       | C       | D       | D            |
| 8               | D       | D       | C       | A       | C       | D       | B       | B       | D            |
| 9               | C       | A       | D       | D       | B       | D       | D       | C       | A            |
| 10              | C       | C       | D       | B       | C       | C       | A       | A       | B            |
| 11              | A       | D       | A       | A       | B       | B       | C       | D       | A            |
| 12              | B       | A       | B       | B       | D       | D       | C       | A       | A            |
| 13              | C       | D       | A       | C       | A       | D       | A       | D       | A            |
| 14              | B       | B       | D       | A       | C       | C       | B       | B       | B            |
| 15              | D       | C       | D       | B       | D       | A       | D       | B       | D            |
| 16              | C       | C       | D       | C       | A       | C       | D       | A       | C            |
| 17              | B       | D       | A       | B       | C       | D       | D       | C       | C            |
| 18              | A       | D       | C       | B       | B       | C       | D       | C       | D            |
| 19              | A       | B       | A       | C       | D       | B       | B       | C       | A            |
| 20              | D       | B       | C       | D       | D       | D       | D       | C       | C            |
| 21              | C       | C       | C       | D       | C       | B       | C       | A       | D            |
| 22              | C       | C       | A       | D       | B       | D       | A       | D       | D            |
| 23              | C       | C       | A       | D       | C       | D       | C       | D       | D            |
| 24              | B       | D       | A       | D       | B       | B       | C       | C       | D            |
| 25              | B       | B       | B       | C       | D       | B       | D       | A       | C            |
| 26              | A       | A       | A       | C       | C       | C       | D       | C       | D            |
| 27              | A       | A       | D       | A       | A       | C       | B       | B       | C            |
| 28              | B       | D       | C       | A       | D       | D       | C       | D       | A            |
| 29              | D       | C       | A       | C       | C       | A       | C       | A       | D            |
| 30              | D       | A       | C       | D       | C       | C       | A       | C       | A            |

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**S**

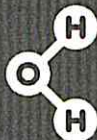
## 2013 Science Answer Keys

**ICAS**  
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| Question Number | Paper A | Paper B | Paper C | Paper D | Paper E | Paper F | Paper G | Paper H | Papers I & J |
|-----------------|---------|---------|---------|---------|---------|---------|---------|---------|--------------|
| 31              |         |         | C       | B       | B       | D       | C       | B       | C            |
| 32              |         |         | A       | C       | C       | C       | B       | C       | B            |
| 33              |         |         | C       | D       | D       | C       | D       | C       | C            |
| 34              |         |         | B       | D       | D       | C       | B       | D       | D            |
| 35              |         |         | D       | D       | D       | C       | B       | C       | C            |
| 36              |         |         | C       | C       | A       | A       | B       | A       | D            |
| 37              |         |         | D       | B       | C       | D       | B       | D       | B            |
| 38              |         |         | B       | D       | C       | C       | A       | B       | C            |
| 39              |         |         | B       | A       | A       | C       | A       | C       | D            |
| 40              |         |         | B       | A       | D       | D       | C       | B       | D            |
| 41              |         |         |         |         | A       | D       | D       | B       | A            |
| 42              |         |         |         |         | A       | D       | C       | B       | A            |
| 43              |         |         |         |         | D       | A       | B       | A       | D            |
| 44              |         |         |         |         | C       | C       | C       | D       | B            |
| 45              |         |         |         |         | B       | B       | D       | D       | A            |

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- Rub out all mistakes completely.
- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.

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|   |   |   |   |   |   |
|---|---|---|---|---|---|
| D | E | B | B | I | E |
|   |   |   |   |   |   |
| A | A | A | A | A | A |
| B | B |   |   | B | E |
| C | C | C | C | C |   |
| D | D | D | D |   |   |
| E |   | E |   |   |   |

**LAST NAME**

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|   |   | B | B |
| C |   | C |   |
| D |   | D |   |
| E |   |   |   |

**FIRST NAME**      **LAST NAME**

CHAN      AIBENG

[illegible][illegible]

☐ Male ☐ Female



☐ Yes ☐ No

**Town / suburb:** \_\_\_\_\_

Postcode: \_\_\_\_\_

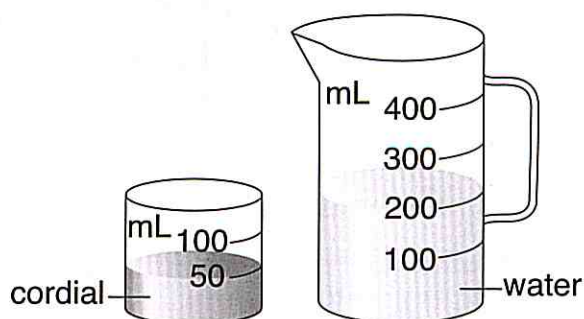
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| 1   | 1 | 1     | 1    | 1 |
| 2   | 2 | 2     | 2    | 2 |
| 3   | 3 | 3     | 3    | 3 |
| 4   |   | 4     | 4    | 4 |
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|  | (F) | (P) |
|  | (G) | (Q) |
|  | (H) | (R) |
|  | (I) | (S) |
|  | (J) | (T) |

## TO ANSWER THE QUESTIONS

**Example:** Ari added cordial to water to make a jug of drink.  
What will be the volume of the drink in the jug?

- (A) 50 mL
- (B) 150 mL
- (C) 200 mL
- (D) 250 mL



The answer is 250 mL, so you would fill in the oval **(D)**, as shown.

(A) (B) (C) **(D)**



## START

- 1 (A) (B) (C) (D)
- 2 (A) (B) (C) (D)
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